

4. [MCQ] [GATE-2017 : 1M]

Match the following according to input from List-I to the compiler phase in the List-II that processes it:

List-I

List-II

- (P) Syntax tree
(Q) Character stream
(R) Intermediate representation
(S) Token stream
- (i) Code generator
(ii) Syntax analyser
(iii) Semantic analyser
(iv) Lexical analyser

- (a) $P \rightarrow (ii), Q \rightarrow (iii), R \rightarrow (iv), S \rightarrow (i)$
(b) $P \rightarrow (ii), Q \rightarrow (i), R \rightarrow (iii), S \rightarrow (iv)$
(c) $P \rightarrow (iii), Q \rightarrow (iv), R \rightarrow (i), S \rightarrow (ii)$
(d) $P \rightarrow (i), Q \rightarrow (iv), R \rightarrow (ii), S \rightarrow (iii)$

Phases of Compiler

[MCQ]

[GATE-2024 : 1M]

Consider the following two sets:

Set X		Set Y	
P.	Lexical Analyzer	1.	Abstract Syntax Tree
Q.	Syntax Analyzer	2.	Token
R.	Intermediate Code Generator	3.	Parse Tree
S.	Code Optimizer	4.	Constant Folding

Which one of the following options is the CORRECT match from Set X to Set Y ?

- (a) P-4: Q-1: R-3 ; S-2 (b) P-2: Q-1: R-3 ; S-4
(c) P-2: Q-3: R-1 ; S-4 (d) P-4: Q-3: R-2 ; S-1

3. [MCQ]

[GATE-2021 : 1M]

Consider the following ANSI C program:

```
int main() {
    Integer x;
    return ();
}
```

Which one of the following phases in a seven-phase C compiler will throw an error?

- (a) Lexical analyzer
(b) Syntax analyzer
(c) Semantic analyzer
(d) Machine dependent optimizer

[MCQ]

[GATE-2023 : 1M]

Consider the following statements regarding the front-end and back-end of a compiler.

- S1: The front-end includes phases that are independent of the target hardware.
S2: The back-end includes phases that are specific to the target hardware.
S3: The back-end includes phases that are specific to the programming language used in the source code.

Identify the CORRECT option.

- (a) Only S1 is TRUE
(b) Only S1 and S2 are TRUE
(c) S1, S2, and S3 are all TRUE
(d) Only S1 and S3 are TRUE

5. [MCQ]

Match the following

List-I

P: Lexical analysis

Q: Top down parsing

R: Semantic analysis

S: Runtime environment

[GATE-2016 : 1M]

List-II

(i) Leftmost derivation

(ii) Type checking

(iii) Regular expressions

(iv) Activation records

- (a) $P \rightarrow (i), Q \rightarrow (ii), R \rightarrow (iv), S \rightarrow (iii)$
(b) $P \rightarrow (iii), Q \rightarrow (i), R \rightarrow (ii), S \rightarrow (iv)$
(c) $P \rightarrow (ii), Q \rightarrow (iii), R \rightarrow (i), S \rightarrow (iv)$
(d) $P \rightarrow (iv), Q \rightarrow (i), R \rightarrow (ii), S \rightarrow (iii)$

Top Down Parser

2. [MSQ]

[GATE-2025 : 2M]

Which of the following statement(s) is/are TRUE while computing First and Follow during top-down parsing by a compiler?

- (a) For a production $A \rightarrow \epsilon$, ϵ will be added to $First(A)$.
(b) If there is any input right end marker, it will be added to $First(S)$, where S is the start symbol.
(c) For a production $A \rightarrow \epsilon$, ϵ will be added to $Follow(A)$.

- (d) If there is any input right end marker, it will be added to $Follow(S)$, where S is the start symbol.

should be 3210)

[MCQ] [GATE-2017 : 1M]

Consider the following grammar:

$$P \rightarrow xQRS$$

$$Q \rightarrow yz \mid z$$

$$R \rightarrow w \mid \epsilon$$

$$S \rightarrow y$$

What is FOLLOW(Q)?

(a) {R} (b) {w}

(c) {w,y} (d) {w,\$}

for next 2 questions)

[NAT] [GATE-2019 : 1M]

Consider the grammar given below:

$$S \rightarrow Aa$$

$$A \rightarrow BD$$

$$B \rightarrow b \mid \epsilon$$

$$D \rightarrow d \mid \epsilon$$

Let a, b, d, and \$ be index as follows:

A	b	d	\$
3	2	1	0

Compute the FOLLOW set of the non-terminal B and write the index values for the symbols in the FOLLOW set in the descending order. (For example, if the FOLLOW set is {a, b, d, \$}, then the answer should be 3210)

[MCQ] [GATE-2024 : 1M]

Which of the following is/are Bottom-Up Parser(s)?

(a) LL(1) Parser

(b) Predictive Parser

(c) Shift-reduce Parser

(d) LR Parser

8. [NAT] [GATE-2021 : 1M]
 Consider the augmented grammar with $\{+, *, (,), id\}$ as the set of terminals.
 $S' \rightarrow S$
 $S \rightarrow S + R \mid R$
 $R \rightarrow R * P \mid P$
 $P \rightarrow (S) \mid id$

lysis
 If I_0 is the set of two LR(0) items $\{[S \rightarrow S.], [S \rightarrow S. + R]\}$, then $goto(closure(I_0), +)$ contains exactly _____ items.

27. [MCQ] [GATE-2010 : 2M]
 The grammar $S \rightarrow aSa \mid bS \mid c$ is
 (a) LL(1) but not LR(1)
 (b) LR(1) but not LL(1)
 (c) Both LL(1) and LR(1)
 (d) Neither LL(1) nor LR(1)

30. [NAT] [GATE-2019 : 2M]
 Consider the augmented grammar given below:
 $S' \rightarrow S$
 $S \rightarrow \langle L \rangle \mid id$
 $L \rightarrow L, S \mid S$
 Let $I_0 = CLOSURE(\{[S' \rightarrow \cdot S]\})$. The number of items in the set $GOTO(I_0, \langle \rangle)$ is: _____.

19. [MCQ]

[GATE-2019 : 1M]

Which one of the following kinds of derivation is used by LR parsers?

- (a) Leftmost
- (b) Leftmost in reverse
- (c) Rightmost
- (d) Rightmost in reverse

21. [MCQ]

[GATE-2015 : 2M]

Consider the following grammar G.

$S \rightarrow F | H$

$F \rightarrow p | c$

$H \rightarrow d | e$

Where S, F and H are non-terminal symbols, p, d and e are terminal symbols. Which of the following statements(s) is/are correct?

S_1 : LL(1) can parse all strings that are generated using grammar G.

S_2 : LR(1) can parse all strings that are generated using grammar G.

- (a) Only S_1
- (b) Only S_2
- (c) Both S_1 and S_2
- (d) Neither S_1 and S_2